



Rozilyn Delenn Marco

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About myself

Robotics researcher focused on the co-optimization of compliant mechanisms and control for contact-rich manipulation. My work combines embedded magnetic sensing and model-based estimation to enable physically grounded external contact inference in tendon-driven robotic systems.

Education & Training

Master of Applied Science (MAsc) in Mechanical Engineering | University of Toronto | 28/08/2025 - 27/08/2027

Continuum Robotics Laboratory (Supervisor: Professor Jessica Burgner-Kahrs)

Level in EQF: 7

Bachelor of Applied Science (BAsc) in Engineering Science | University of Toronto | 31/08/2019 - 14/06/2025

Final grade: 3.51/4.00 (With Honours) | Level in EQF: 6 | Thesis: An Open-Source, Extensible, Tendon-Driven Continuum Robotic System

Publications

Automated Planning Domain Inference for Task and Motion Planning

Jinbang Huang, Allen Tao, **Rozilyn Marco**, Miroslav Bogdanovic, Jonathan Kelly, Florian Shkurti

International Conference on Robotics and Automation (ICRA), 2025

Robotic Food Handling Utilizing Temperature Dependent Variable-Stiffness Material

Rozilyn Marco, Prashant Kumar, Xinyi Zhang, Weiwei Wan, Kensuke Harada

International Conference on Artificial Life and Robotics (ICAROB), 2024

Work experience

Graduate Researcher with Professor Jessica Burgner-Kahrs | Continuum Robotics Laboratory, University of Toronto Mississauga | 31/07/2025 - Current | Canada

- Conduct research on the design, modeling, and control of tendon-driven continuum robots.
- Develop mechanical prototypes using multi-material 3D printing and performs experimental characterization of curvature, force–displacement behaviour, and workspace metrics.
- Contributes to an open-source continuum robotics platform through validated fabrication processes and reproducible testing procedures.

Research Intern with Professor Jonathan Kelly | STARS Laboratory, University of Toronto | 31/03/2024 - 30/06/2024 | Canada

- Contributed to a deep learning framework integrating Graph Attention Networks with automated planning for task-domain inference.
- Developed and validated an RGB-D pose estimation pipeline for extracting task conditions for PDDL generation.
- Evaluated learned planning domains on a Franka Research 3 robot for multi-step manipulation tasks.
- Co-authored a publication accepted to ICRA 2025 and delivered the spotlight presentation and poster session.

Robotic Research Intern with Professor Kensuke Harada | Robotic Manipulation Lab, Osaka University | 30/03/2023 - 29/06/2023 | Japan

- Designed and prototyped a variable-stiffness end-effector using deformable materials.
- Conducted mechanical characterization experiments involving temperature-dependent stiffness changes and force-gauge measurements.

- Refined designs based on experimental results to improve force transmission and stability.
- Work led to a first-author publication at ICAROB 2024.

Data Analyst with Professor Emily Moore | Institute for Studies in Transdisciplinary Engineering, University of Toronto | 30/07/2022 - Current

- Designed and maintained engagement databases (2,400+ records), improving data quality and traceability across internal systems.
- Implemented automated data-processing pipelines in R and developed analytical dashboards (Excel, Power BI) to support program evaluation and evidence-based decision-making.

Honours and Awards

NSERC Undergraduate Student Research Award (USRA) | Natural Sciences and Engineering Research Council of Canada (NSERC) | 01/05/2024

- Competitive national award funding a full-time 16-week undergraduate research project.

First Place - International IEEE Solid-State Circuits Society Arduino Contest | Institute of Electrical and Electronics Engineers (IEEE) | 01/09/2023

- Awarded for technical excellence and innovation in hardware–software system design.

Engineering Science Research Opportunities Program (ESROP) - Global | University of Toronto | 01/05/2023

- Competitive fellowship supporting an international summer research placement.

FrontierLab Mini | Osaka University | 01/05/2023

- Award supporting a short-term international research placement.

Dean's Merit Award | University of Toronto | 11/10/2019

- Awarded for academic excellence, recognizing high-achieving incoming engineering students with outstanding admission averages.

Projects

Open Source Tendon-Driven Continuum Robot (TDCR) | 29/07/2024 - 26/02/2025

- Designed and fabricated an extensible TDCR with magnetically self-distributing spacer disks.
- Implemented real-time teleoperation for combined bending and extension.
- Developed experimental protocols for workspace, curvature, and follow-the-leader performance evaluation.

https://github.com/rozmarco/extensible_tdcr

Autonomous Agricultural Drone | 30/11/2024 - 28/02/2025

- Designed perception and mechanical mounting systems for a quadcopter used in precision agriculture.
- Integrated ROS/MAVROS with custom-trained object detection models for navigation and crop detection.
- Collaborated across mechanical, perception, and planning teams to deliver a fully autonomous pipeline.

https://github.com/CathyF9600/capstone_ws

Conferences & Seminars

International Conference on Robotics and Automation (ICRA) | 13/05/2025 - 17/05/2025 | Atlanta, United States

- Presented co-authored, published work during oral and poster sessions.

Montréal Robotics Summer School with Professor Glen Berseth | 02/08/2025 - 07/08/2025 | Montréal, Canada

- Selected for fully funded intensive program in manipulation, motion planning, RL, and perception.
- Team achieved top forward-walking performance in quadruped locomotion challenge.
- Hosted by Mila - Quebec Artificial Intelligence Institute.

International Conference on Artificial Life and Robotics (ICAROB2024) | 02/2024 | Oita, Japan

Skills

Continuum robot modeling | Optimization (continuous, discrete, nonlinear, mixed-integer) | Motion planning | Reinforcement learning (RL) | ROS/ROS2 | MuJoCo | Gazebo | Python (NumPy, SciPy, PyTorch) | MATLAB | C | CAD (Fusion 360) | Multi-material 3D printing | Git | Linux

Industry Experience

Junior Software Developer | Linux Professional Institute | 26/04/2021 - 27/08/2022

- Developed backend automation tools using PHP, Laravel, and Bash to support digital credential and exam generation workflows.
- Built utilities for automated image and JSON file creation and developed Grafana dashboards using SQL-based analytics.
- Updated internal codebases and documentation to support technical teams across deployment and reporting processes.

Robotics Technician | Brand M3dia Group | 31/08/2021 - 14/07/2022

- Operated and programmed the company's AI robots, doubling user engagement over two months.
- Produced, choreographed, and filmed skits used for company marketing and custom event entertainment.
- Wrote current robot operating manuals and training tools now used as the company's standard.

Software Developer Intern | Linux Professional Institute | 04/05/2020 - 28/08/2020

- Implemented API endpoints and backend features using PHP and the Laravel framework.
- Developed tools for generating exam materials in PDF and XML formats.
- Collaborated with technical staff to update internal codebases and enhance reliability of exam delivery workflows.

Volunteering

Robotic Competition Judge | FIRST Tech Challenge | 30/09/2025

- Evaluated robotics teams on technical design, system integration, and communication.
- Participated in award deliberations and technical interviews.

Engineering Mentor | Engineers Without Borders, University of Toronto Chapter | 28/11/2019 - 29/01/2021

- Mentored high-school teams in engineering design competitions and community-impact programs.
- Guided students through problem formulation, design trade-offs, and prototype refinement.